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Observations on the Vibrations of Piano Strings *

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The behavior of single piano strings struck in their normal positions was investigated through measurement of the frequencies of the partials and their amplitudes as functions of time after striking. Three-dimensional figures showing partial amplitudes *versus* time are presented. The frequencies of the partials of the tone were found to be those of the normal modes of vibration and are therefore referred to as the modal frequencies of the string. The higher modal frequencies are progressively sharp with respect to true harmonics of the fundamental; the sharpening follows approximately a square law with respect to mode number, to the extent that in one case the 15th partial had just about the frequency of a 16th harmonic. The resulting inharmonicity in piano strings is least in the two octaves below middle C; it rises sharply at the high end and more or less sharply at the low end, depending upon the type of piano. A correlation between inharmonicity and subjective tone quality rating is suggested. It is shown that the commonly observed "stretched" tuning (sharpening of the treble, flattening the bass) is a natural consequence of the inharmonicity.

INTRODUCTION

INVESTIGATION of the tone and vibration of the piano string, when strung in the piano and struck by the hammer in normal fashion, is complicated by the transitory nature of the phenomenon. However, it is possible to make such studies by separating out one partial of the tone at a time with a selective filter. This partial's amplitude variation with time after striking may then be recorded with a high speed graphic level recorder and its frequency may be determined with a chromatic stroboscope. These

data, taken for each of the distinguishable partials in the tone, indicate the nature of the tone quality and furnish information regarding the vibration of the string. Some observations of this kind are described below, and their significance is discussed, both in relation to tone quality and the tuning of pianos.

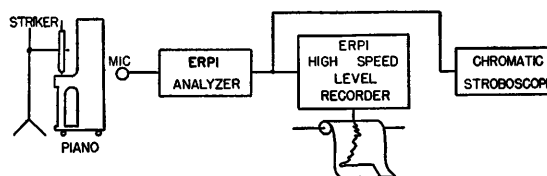
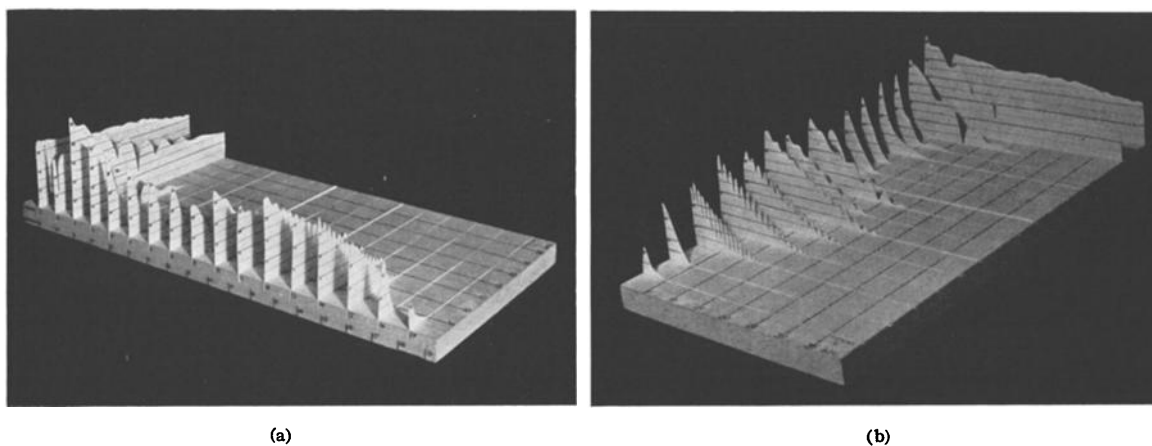
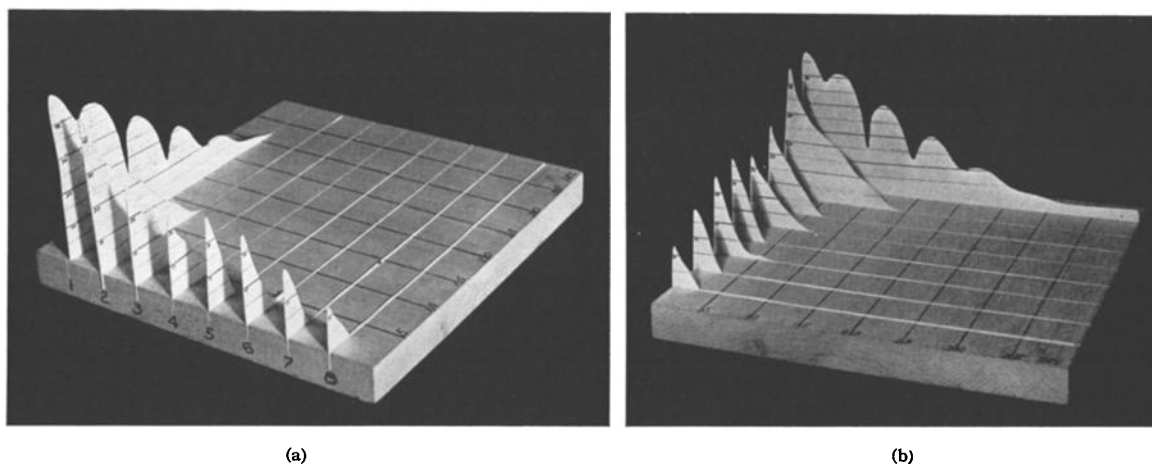


FIG. 1. Apparatus for measuring amplitudes of partials.

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FIG. 2. Amplitudes of the partials of an F_1 string as functions of time.FIG. 3. Amplitudes of the partials of an F_3 string as functions of time.

AMPLITUDE OF PARTIALS VERSUS TIME

Figure 1 shows the arrangement of apparatus for measuring the variation of amplitude with time after striking, for each partial of a piano tone. The selected key is struck by a mechanical striker; when the key has two or three strings, only one is left free to vibrate. The sound is picked up by a microphone, fed through an analyzer which is tuned to one partial; the amplitude of this partial is recorded by the high speed level recorder, while its frequency is read on the chromatic stroboscope. A record is run for each of the ascending partials until they can no longer be distinguished. The records are similar in form to the examples obtained by Wolf and Sette.¹

¹S. K. Wolf and W. J. Sette, J. Acous. Soc. Am. 6, 160-168 (1935).

In order to present the results in a form readily comprehended, fiber templets were cut from the level recorder records and mounted on slotted boards to form three-dimensional figures, as shown in Figs. 2 and 3. Figure 2 represents the tone from the F_1 string—the lowest F —on a small console piano. Similarly, Fig. 3 represents the tone from one string of the key F_3 —the third F , the F below middle C —on the same piano. The ordinates are on a logarithmic scale.

It is seen that after the initial peak at striking,

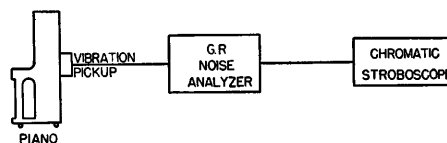


FIG. 4. Method for determining modal frequencies of piano strings.

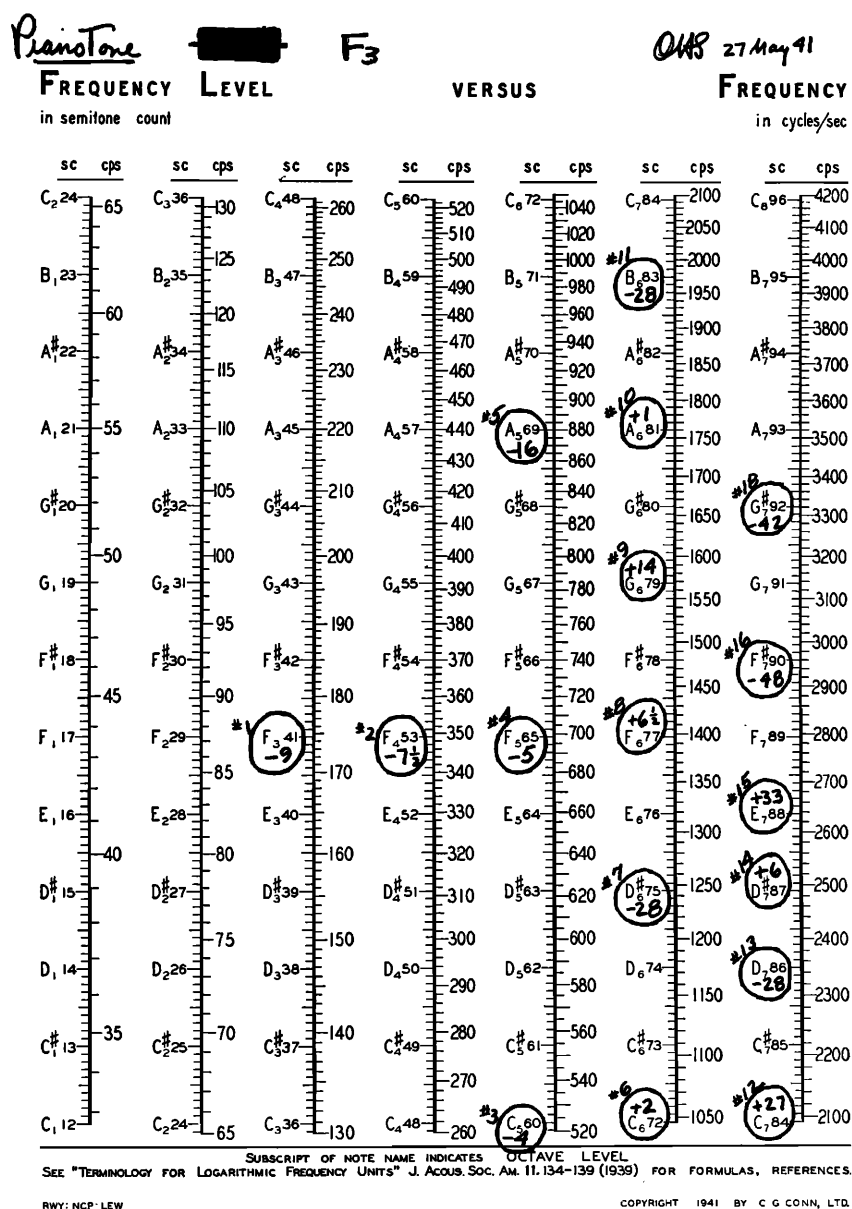


FIG. 5. Frequency level conversion chart used as a data sheet.

which is more rapid in its rise than the high speed level recorder can follow (the nominal maximum being 50 db/sec.), there is a fairly uniform rate of decrease in amplitude, the rate of decrease being more rapid the higher the number of the partial. This is to be expected from the increased air loading of the string and sounding board at the higher frequencies. The tone quality therefore changes continuously, becoming less

complex as time goes on.² Particularly in the tone produced by the longer string (Fig. 2) the amplitudes of the partials show large fluctuations of a form suggesting half sine waves. The higher the partial, the more rapid the fluctuations. This condition may be due to rotation of the plane of vibration of the string, and there

² Hart, Fuller, and Lusby, J. Acous. Soc. Am. 6, 92 (1934).

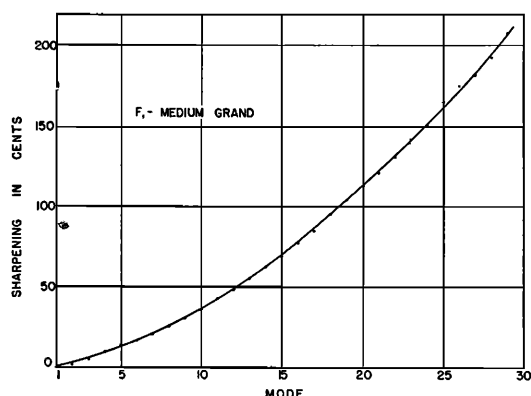


FIG. 6. Inharmonicity for the F_1 string of a medium grand piano.

may also be some transfer of energy from one mode to another. It is interesting to notice that the seventh partial is not particularly weak, despite the fact that the striking point is often stated to be at one-seventh of the string length to suppress this partial.

The frequency scale used in making the three-dimensional figures is linear: Harmonics would therefore appear equally spaced. The templates have been located to correspond to the measured frequencies. As is particularly evident in Fig. 2a, the frequencies of the partials are not harmonic. Along the edge of the board the number of the mode of vibration which produces each partial is indicated in the upper row of figures; in the lower row the position of each true harmonic is shown. It is seen that the partials are progressively sharpened, indeed to the extent that the 15th partial has just about the frequency of a 16th harmonic. True harmonics of the fundamental frequency could not be found. This fact indicates that the transmission system from string to air is relatively distortionless, and that the partials in the tone are indeed produced by the respective modes of vibration of the string.

It should be pointed out that these amplitude data are unquestionably influenced by the room conditions, and are only to be considered as indicating the practicability of making measurements of this kind. On the other hand, the frequencies of the partials are not influenced by the room conditions. Their measurement is relatively simple and since their influence on tone quality was considered important, all of the further

work described here was concerned only with such frequency measurements.

FREQUENCIES OF PARTIALS

The frequencies of the various partials of a struck piano string were measured by the apparatus shown in Fig. 4. (Since these frequencies are those of the normal modes of vibration they are hereafter referred to as the modal frequencies.) The output of a crystal vibration pick-up fastened to a rib of the sounding board was fed through an analyzer to a chromatic stroboscope. Only one string of the group being struck was left free to vibrate. As it was found that variations in the strength of the blow produced negligible effects on the modal frequencies, the key was struck manually.

Rapid recording of the data was facilitated by use of the frequency level *versus* frequency chart shown in Fig. 5, this being the actual work sheet of one run. The number of cents

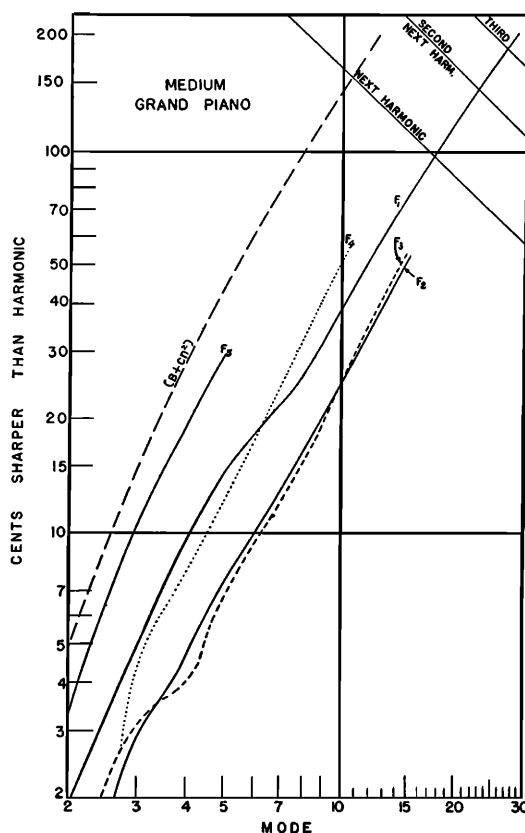


FIG. 7. Inharmonicity as some power of mode number.

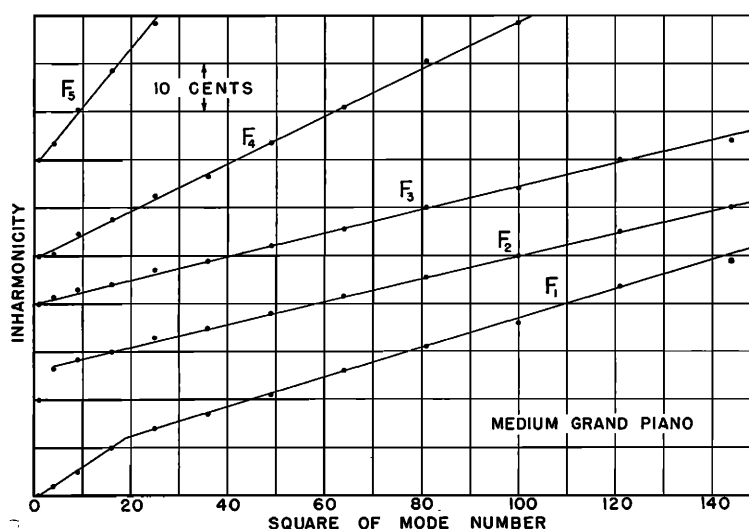


FIG. 8. Inharmonicity as a function of the square of mode number.

deviation measured with respect to a nominal value is encircled; the corresponding mode number is indicated alongside.

The readings are in terms of frequency level,^{3,4} so the percentage sharpening of each partial with respect to a true harmonic is easily obtained by a subtraction procedure which is described in the Appendix. We call this percentage departure from harmonic the "inharmonicity." More specifically it is convenient to measure and express the inharmonicity as the number of *cents* by which a particular modal frequency ν departs from its nominal value $n\nu_1$, where ν_1 is the fundamental frequency and n is the mode number. This departure, the inharmonicity D , is therefore

$$D = 1200 \log_2 (\nu / n\nu_1). \quad (1)$$

It may be recalled that the cent is one hundredth of a semitone, so there are 1200 to the octave; one cent corresponds to a percentage difference in frequency of 0.06 percent. D is zero if the modal frequency is exactly harmonic.

The inharmonicities of the modal frequencies of one string in a medium grand piano are shown in Fig. 6, plotted against the number of the mode. Note that the inharmonicity follows a fairly smooth curve and is always positive. This means that the partials are always sharp

with respect to true harmonics. When the inharmonicity *versus* mode number is plotted on log-log paper, as shown in Fig. 7, the observed points for a number of strings measured in the medium grand piano fall, respectively, on fairly straight lines whose slopes are close to 2. This means that approximately

$$D = 1200 \log_2 \nu / n\nu_1 \approx Kn^2, \quad (2)$$

where K is some factor of proportionality. This is nearly equivalent to

$$\nu = n\nu_1 [1 + (K/1731)n^2], \quad (3)$$

as may be shown by substitution back in Eq. (2) and using the appropriate expansion for the logarithm.

This proportionality to n^2 is of the form predicted by theory for stiff strings and various boundary conditions,⁵⁻⁷ provided that the terms independent of mode number are absorbed in ν_1 .

The line in Fig. 7 labeled "next harmonic" indicates at what point a particular mode has been sharpened to the extent that its frequency is equal to that of a true harmonic one number greater. Similarly, the intersection of the curve for F_1 near $n=23$ with the line "second next

⁵ Lord Rayleigh, *Theory of Sound* (Macmillan, 1894), second edition, Vol. I, p. 301.

⁶ R. S. Shankland and J. W. Coltman, *J. Acous. Soc. Am.* **10**, 161-166 (1939).

⁷ P. M. Morse, *Vibration and Sound* (McGraw-Hill, 1936), p. 131.

³ H. Fletcher, *J. Acous. Soc. Am.* **6**, 59-69 (1934).

⁴ R. W. Young, *J. Acous. Soc. Am.* **11**, 134-139 (1939).

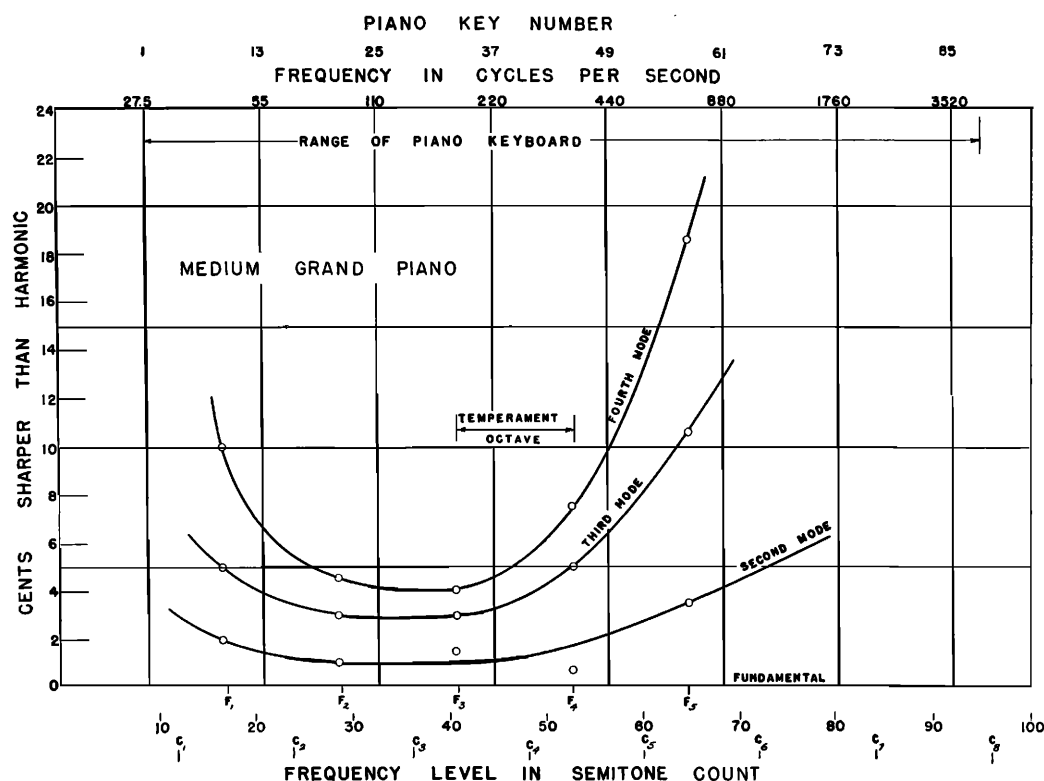


FIG. 9. Inharmonicity in relation to position of string in piano.

harmonic" means that the frequency of this mode of vibration is approximately 25 times that of the fundamental.

In order to investigate the "square" relationship in detail the inharmonicity *versus* square of mode number has been plotted in Fig. 8, again using the data on the medium grand piano. The curves have been given an arbitrary vertical displacement for clarity. The straight lines are for convenience in visualizing the relationship; actually only discrete values of the inharmonicity exist corresponding to the mode numbers.

For the higher pitched strings a single straight line constitutes an excellent fit. For the lower ones a constant should be added to the second member of Eq. (2), and the exponent reduced to something less than 2. The effect of such an additive constant is illustrated by the curvature of the dashed line in Fig. 7 labeled $(B + Cn^2)$, where B and C have been chosen arbitrarily. Such a complication should perhaps be expected for F_1 , in consideration of yielding of the end supports and the heavy loading.

COMPARISON OF STRINGS

From Figs. 7 and 8 it is apparent that the inharmonicity is not the same for all strings. In Fig. 9 the inharmonicity for several of the lower modes is replotted against frequency level and position on the piano keyboard. The sharpening is least in the two octaves below middle C and increases rapidly below and above this region. In fact, it shows signs of reaching quite surprising values at the top end of the keyboard. This is quite plausible for the top strings are so short and stiff as to be really more like bars than strings.

One would surmise that, since all pianos have about the same string dimensions in the middle and upper registers, there would be little variation here in the inharmonicity among different pianos. On the other hand, the lengths of the bass strings vary with the size of the piano, and one would accordingly expect variations in the inharmonicity in the low register. These expectations were corroborated by measurements on different sizes of pianos.

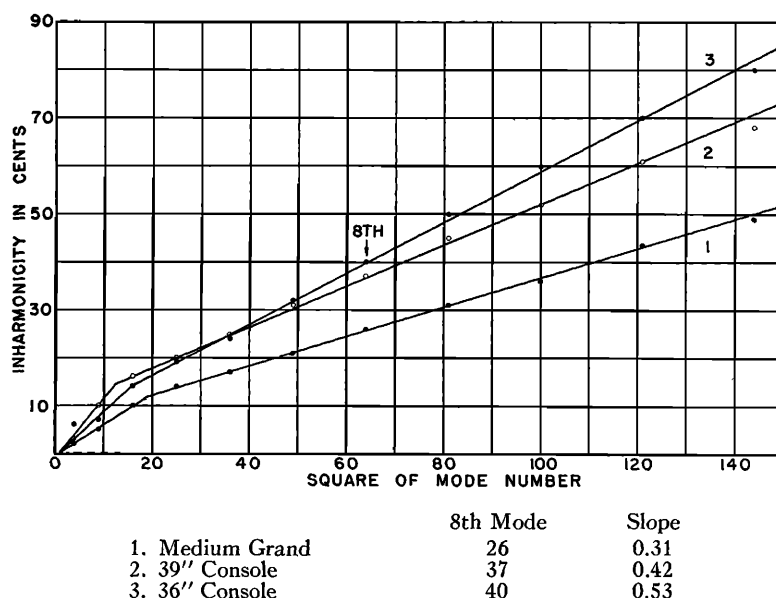


FIG. 10. Comparison of the inharmonicity of F_1 for various pianos.

Figure 10 gives a comparison of the inharmonicity of the F_1 strings of several pianos, again plotted against the square of the mode number for convenience. The inharmonicity is less for the larger piano. The eighth modes, for example, were respectively 26, 37, and 40 cents sharper than the corresponding harmonic values. This observation is in line with the widely held opinion that the piano with longer bass strings has a "rounder, fuller, richer tone." Let us consider this point philosophically for a moment.

It is generally agreed that a tone is round, full, and rich if it consists of a fundamental and a long series of harmonic overtones of gradually decreasing amplitudes. Nevertheless, if the fundamental is missing or weak, as it often is in the lower tones of a piano, the listener will still imagine he hears a fundamental in the series and be pleased with the tone. This subjective fundamental is derived from the differences between the successive partials. When they are harmonically related as when produced, for example, by a horn, the frequency differences between successive partials are constant, and the derived fundamentals all have the same frequency, which is that of the actual fundamental. However, if the partials are not harmonically related, as in the piano tone, the various derived fundamentals do not have the same

frequency, there is confusion in the ear, and the tone is not considered pleasing.

The inharmonicity of some arbitrarily selected mode therefore constitutes a simple criterion of excellence of a string. However, particularly for bass strings, the fundamental is often almost lacking, so the accurate determination of the interval from the fundamental to the eighth mode, say, is not always easy. Moreover, the manner in which the inharmonicity changes with mode number may vary among different pianos. On the hypothesis suggested above it is the differences between successive partials which are significant, so that the rate of increase of inharmonicity with mode number is pertinent. The caption with Fig. 10 shows also this method of rating, the slope relative to n^2 having been used for simplicity. Observe that although pianos 2 and 3 had nearly the same inharmonicity of the eighth mode, the greater slope of the curve 3 indicates a greater spread of differences between successive partials. This criterion of slope relative to n^2 may be determined experimentally with accuracy in a straightforward manner. On the other hand, there may be needed some power of mode number other than two, and the criterion does not recognize that the subjective fundamental created within the ear will depend upon the actual frequencies and



FIG. 11. (a) Left: Calculated tunings, showing deviations in cents.
(b) Right: Average of 16 pianos tested by Railsback.

relative amplitudes of the various partials involved.

That the longer string is desirable for its better tone quality is appreciated by the piano designer,⁸ who takes considerable pains to make the bass strings as long as the dimensions of the case will allow.

EFFECT OF INHARMONICITY ON TUNING

There is another effect of interest produced by the inharmonicity of the partials. Measure-

ments by Railsback⁹ show that piano tuners usually tune progressively sharp in the middle and high registers, progressively flat in the low. The process of piano tuning^{10,11} is carried out by "laying the temperament" in an octave in the middle of the keyboard—usually the octave from F_3 to F_4 —and then tuning up and down by octaves. To tune upward by an octave, one

⁹ O. L. Railsback, *J. Acous. Soc. Am.* **9**, 274 (1938); **10**, 86 (1938).

¹⁰ G. F. H. Harker, *J. Acous. Soc. Am.* **8**, 243–256 (1937).

¹¹ W. B. White, *J. Acous. Soc. Am.* **9**, 47–50 (1938).

⁸ W. B. White, *J. Acous. Soc. Am.* **12**, 409–411 (1941).

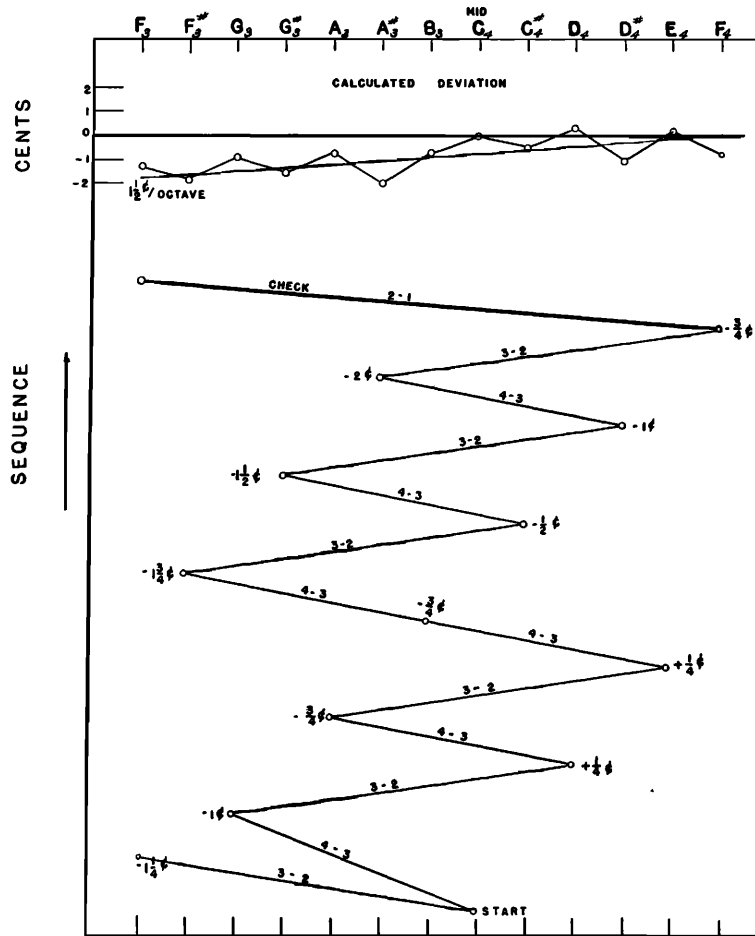


FIG. 12. Deviations from equal temperament within an octave resulting from inharmonicity.

adjusts the tension of the upper string until its fundamental has the same frequency as the second mode of the lower string, as indicated by the absence of beats. If the second mode of the lower string is sharper than a true second harmonic of the fundamental, then the upper string's fundamental will be tuned sharp. The next octave jump will increase the sharpening still further, particularly since the inharmonicity increases toward the high end, as shown in Fig. 9. From the curve there given for the second mode one can derive the expected tuning curve for this particular piano as given by (2-1) in Fig. 11, in which deviations from nominal equal temperament are given in cents. It is seen to be very similar in shape to the average tuning curve found by Railsback (right-hand curve), although its deviations are not as great as he shows.

If the fundamental is missing or weak, as is often the case in the lower register, it may be that rather than beating first and second, the tuner is actually adjusting the second partial of the upper tone to have the same frequency as the fourth partial of the lower tone. When this is the case the tuning curve marked (4-2) in Fig. 11 should be obtained. It has the same shape as (2-1), but its deviations are much greater, even greater than those of Railsback's average curve in the upper register. However, it is quite comparable in the lower register where this method of tuning is more likely to be used.

A tuner often checks his tuning by double octaves, that is, beating the fourth partial of the lower tone against the fundamental of the upper tone. If this were done as a tuning procedure, the resulting tuning curve would be that labeled

(4-1) in Fig. 11, which is seen to fall between the curves. Which of these methods of tuning is actually used, and in which section of the keyboard, appears to depend upon the individual tuner and the particular piano he is tuning.

In laying the temperament initially the tuning is also affected somewhat by the sharpening of the partials. The tuner adjusts the fundamental of one string to his tuning fork, then adjusts another string to the first by listening for beats

TABLE I. Semitones corresponding to numbers to 50.
 $\text{Semitones} = \log_2 \frac{1}{N}$ $N = 12 \times \log_2 N$.
 100 cents = 1 semitone.

N	Interval	Semitones	Cents	N	Interval	Semitones	Cents
1	12.00	0.00	0.00	26	0.65	56.41	5640.53
2	7.02	12.00	1200.00	27	.63	57.06	5705.87
3	4.98	19.02	1901.95	28	.61	57.69	5768.82
4	3.86	24.00	2400.00	29	.59	58.30	5829.58
5	3.16	27.86	2786.31	30	.57	58.88	5888.27
6	2.67	31.02	3101.95	31	.55	59.45	5945.04
7	2.31	33.69	3368.82	32	.53	60.00	6000.00
8	2.04	36.00	3600.00	33	.52	60.53	6053.27
9	1.82	38.04	3803.91	34	.50	61.05	6104.95
10	1.65	39.86	3986.31	35	.49	61.55	6155.14
11	1.51	41.51	4151.32	36	.47	62.04	6203.91
12	1.39	43.02	4301.95	37	.46	62.51	6251.34
13	1.28	44.41	4440.53	38	.45	62.98	6297.51
14	1.19	45.69	4568.82	39	.44	63.42	6342.48
15	1.12	46.88	4688.27	40	.43	63.86	6386.31
16	1.05	48.00	4800.00	41	.42	64.29	6429.06
17	0.99	49.05	4904.95	42	.41	64.71	6470.78
18	0.94	50.04	5003.91	43	.40	65.12	6511.51
19	0.89	50.98	5097.51	44	.39	65.51	6551.32
20	0.84	51.86	5186.31	45	.38	65.90	6590.22
21	0.81	52.71	5270.78	46	.37	66.28	6628.27
22	0.77	53.51	5351.32	47	.36	66.66	6665.51
23	0.74	54.28	5428.27	48	.36	67.02	6701.95
24	0.71	55.02	5501.95	49	.35	67.38	6737.65
25	0.68	55.73	5572.63	50		67.73	6772.63

TABLE II.

1 n	2 sc	3 sem	4 sc	5 cents
1	40.91	0.00	40.91	
2	52.92	12.00	40.92	1
3	59.96	19.02	40.94	3
4	64.95	24.00	40.95	4
5	68.84	27.86	40.98	7
6	72.02	31.02	41.00	9
7	74.72	33.69	41.03	12
8	77.06	36.00	41.06	15

Column headings: 1. Mode number; 2. Frequency level; 3. Nominal harmonic interval; 4. Frequency level of "reduced" fundamental; 5. Inharmonicity.

between partials of the same approximate frequency. He adjusts the beats to be in accordance with a table calculated on the assumption that the partials are truly harmonic. In Fig. 12 is illustrated the sequence used by one tuner. Starting at the bottom with C_4 (middle C), the lines indicate the intervals being adjusted, each being labeled with the partials which are beating. Taking the corresponding inharmonicities from Fig. 9 and subtracting gives the amounts in cents by which the intervals are long. The resulting tuning is plotted at the top of the figure and is seen to be somewhat irregular, but the general rise is quite similar to that often found throughout the middle register of the piano.

The measurements described here indicate that the "stretched"¹¹ tuning is a natural result of the inharmonicity of the partials of the piano tone and is perhaps necessary if the piano is to sound in tune with itself. Whether or not stretching is desirable may therefore depend upon the size and construction of the piano and whether it is to be used alone or with other instruments.

CONCLUSION

The observations described above, while admittedly preliminary, show definitely that the actual strings used in the piano do not behave like idealized strings. Their modal frequencies do not follow exactly the harmonic series, but are progressively sharp with respect to true harmonics. At least within certain limits the inharmonicity is approximately proportional to the square of the mode number, and increases considerably toward the upper and lower ends of the keyboard. It seems to be about the same

in the middle and upper registers of different pianos, but to be less in the lower registers of pianos with longer bass strings. These observations lead to a plausible explanation of the better tone quality generally ascribed to pianos with longer bass strings.

A consideration of piano tuning procedure indicates that the effect of the inharmonicity is to produce just the sort of deviations in tuning that have actually been observed. Much work remains to be done to add to the data here presented, to study in detail the effects of bridge forms, braces, and string construction, and to investigate further the influence on tuning.

The authors are indebted to O. M. Salati and O. H. Kraushaar for their cooperation in making the measurements.

APPENDIX

Routine for Establishing Inharmonicity

The basis of this method is the comparison of the logarithms of the ratios of the modal frequencies with the logarithms of the nominal ratios. Furthermore, by choosing as the base of logarithms $2^{1/12}$, the ratios are measured

directly in semitones (or cents), and are thus capable of convenient interpretation in terms of the musical intervals.

Table I gives these logarithms for integers to 50. This represents an expansion of a similar table published by Wead¹² and a few corrections have been made. The table shows, for example, that a true seventh harmonic is 33.69 semitones above the fundamental, and that the interval between the seventh and eighth harmonics is 2.31 semitones.

Table II illustrates the subtraction process, based on the data given in Fig. 5. The eighth mode, for example, was found to have a frequency level of 77.06 sc (semitones counted from a reference frequency of 16.35 cycles/sec.). Upon subtracting 36.00 from this value we obtain the frequency level of the fundamental, had this mode been harmonic. Then by subtracting the frequency level of the measured fundamental we arrive at the number of semitones (rather with the decimal point moved over for cents) by which each mode exceeds its harmonic value. The last step is unnecessary in constructing curves as in Fig. 10, since the frequency level of the "reduced fundamentals" may be plotted directly. This is a convenience when an accurate measurement of the fundamental frequency is not available.

The position at which to match the nominal intervals is found by inspection. If the differences in column 4 of Table II are not approximately constant a poor choice of the starting point has been made.

¹² C. K. Wead, Report of the National Museum for 1900, U. S. Government Printing Office (1902), pp. 417-462.